Muhammad Uzair

Electrical Engineer | Embedded Systems | Industrial Automation

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PROFESSIONAL SUMMARY

Electrical Engineering graduate (FAST-NUCES, 2022-2026) with hands-on experience in instrumentation, control systems, embedded electronics, IoT, and industrial automation. Worked on academic and real-world projects involving microcontrollers, simulation tools, PLC programming, and ML applications.

EDUCATION

Bachelor of Electrical Engineering
FAST-NUCES | Chiniot-Faisalabad Campus
2022 - 2026 Chiniot-Faisalabad, Pakistan
Studied control systems, instrumentation, embedded systems, power electronics and DSP.
Active IEEE member; participated in project exhibitions and engineering competitions.

F.Sc Pre-Engineering
Government Post Graduate College, Sahiwal
2020 - 2022 Sahiwal, Punjab, Pakistan
Focus: Physics, Chemistry, Mathematics

PROFESSIONAL EXPERIENCE

Electrical Intern
Shaukat Khanum Memorial Cancer Hospital and Research Centre
July 2025 | 4 Weeks | Lahore, Pakistan
- Assisted in maintenance and troubleshooting of laboratory machines and electrical systems.
- Practical exposure to instrumentation, preventive maintenance and calibration procedures.
- Applied laboratory safety standards and equipment handling protocols.

Electrical and Instrumentation Intern
Ibrahim Fiber Limited
June 2025 | 4 Weeks | Faisalabad, Pakistan
- Assisted in installation, calibration and maintenance of industrial instruments.
- Practical exposure to control systems, sensors and signal transmitters.
- Supported engineers in monitoring automated production line processes.

KEY PROJECTS

Intelligent Fault-Tolerant Sensor Assembly (IFTSA) - Final Year Project
ML-based smart sensor system that detects, isolates and compensates sensor faults in real-time.
Tools: MATLAB, Jupyter Notebook, Proteus | Skills: ML, redundant sensors, embedded control

IoT Water Level Monitor
Remote water-level tracking via ESP32 + HC-SR04 ultrasonic sensor, streamed over WiFi/MQTT.
Tools: Arduino IDE | Skills: IoT, ESP32, wireless telemetry, real-time dashboards

DC-DC Boost Converter with Microcontroller Control
High-efficiency boost converter with MCU feedback loop and PWM regulation, simulated in Proteus.
Tools: Proteus, Multisim, MATLAB | Skills: Power electronics, analog circuits

Traffic Light Control System using PLC and HMI
Automated traffic junction controller using PLC ladder logic and HMI display in TIA Portal.
Tools: Siemens TIA Portal | Skills: PLC, SCADA, HMI, industrial automation

TECHNICAL SKILLS

- Embedded Systems: Arduino, ESP32, Raspberry Pi
- PLC Programming: TIA Portal, Ladder Logic, SCADA
- Circuit Design: Proteus, Multisim, PCB Prototyping
- Power Electronics: Boost/Buck converters, PWM
- IoT: WiFi, MQTT, Sensor Networks
- Tools: MATLAB, AutoCAD, Vivado, LabVIEW, VS Code
- Machine Learning: Python, Jupyter Notebook, scikit-learn
- Soft Skills: Documentation, teamwork, problem-solving

CERTIFICATIONS AND ACHIEVEMENTS

- Instrumentation and Control | Industrial Automation Control - Inst Tools
- IEEE Executive Member | IEEE Student Branch, FAST-CFD
- Speed Wiring Competition Winner | IEEE Speed Wiring Competition 2024